LearnHub

E-Learning Mobile Application

Document 1: Problem Statement, Business Case, Scope & Project Goals

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Mobile Application Development Course

1. Problem Statement

In today's fast-paced digital world, learners face significant barriers when accessing quality educational content. Traditional learning platforms often require constant internet connectivity, making them inaccessible in areas with limited or unstable network coverage. Additionally, many existing e-learning applications are overly complex, lack personalization features, and fail to provide meaningful progress-tracking mechanisms that motivate learners to continue their studies.

Students and self-learners frequently struggle with the following challenges:

- Dependency on internet connectivity to access course materials and track progress.
- Lack of offline data persistence, causing users to lose their enrollment status and learning progress.
- Poor personalization — most platforms do not allow users to customize their interface or preferences.
- Absence of an intuitive progress-tracking system that clearly shows learning advancement per course.
- Overly complicated navigation that discourages non-technical users from adopting the platform.

These problems collectively reduce learner engagement, decrease course completion rates, and limit the overall effectiveness of digital education tools.

2. Business Case

LearnHub addresses a growing market demand for accessible, offline-capable e-learning solutions. The global e-learning market is experiencing rapid expansion, yet a significant portion of potential learners — particularly those in developing regions or with limited data plans — are excluded due to connectivity requirements.

2.1 Value Proposition

LearnHub delivers a lightweight, fully offline mobile application built with Flutter and Dart that works seamlessly on both Android and iOS platforms. By storing all user data locally via SharedPreferences, the application eliminates the need for a backend server or internet connection, significantly reducing infrastructure costs while maximizing accessibility.

2.2 Target Audience

- University and college students seeking structured self-learning tools.
- Working professionals pursuing skill development in programming, design, and business.
- Learners in regions with unreliable or expensive internet access.

- Instructors and institutions looking for a simple course delivery platform.

2.3 Business Benefits

Benefit	Description
Zero Server Cost	All data stored locally; no backend infrastructure required.
Cross-Platform	Single Flutter codebase deploys to Android and iOS simultaneously.
Offline Access	Full functionality without internet connectivity.
High Retention	Interactive sliders and progress tracking encourage continued usage.
Scalability	Easy to extend with new courses, categories, and features.

3. Project Scope

3.1 In Scope

- A Flutter-based mobile application targeting Android and iOS platforms.
- Splash screen with a 3-step onboarding flow shown on first launch only.
- Home screen displaying 8 courses with category filter chips and real-time search.
- Course detail screen with description, lesson count, rating, and enroll/unenroll functionality.
- Enrollment system with local persistence using SharedPreferences.
- Interactive progress slider (0–100%) per course with stored completion percentage.
- My Courses screen showing enrolled courses, average progress, and completed count.
- Profile & Settings screen with username editing, dark mode toggle, and data reset.
- Full dark/light theme support applied app-wide via ThemeMode.
- Form validation for profile name input.

3.2 Out of Scope

- Real-time video streaming or downloadable course content.
- Backend server, cloud database, or user authentication system.
- Payment processing or subscription management.
- Social features such as discussion forums, peer reviews, or messaging.
- Push notifications or scheduled reminders.
- Administrative dashboard for course management.

4. Project Goals

4.1 Primary Goals

1. Deliver a fully functional offline e-learning mobile app using Flutter & Dart.
2. Implement local data persistence for enrollments, progress, theme, and username.
3. Provide a smooth, intuitive user experience with proper navigation and state management.
4. Enable full dark/light mode personalization stored across sessions.
5. Ensure cross-platform compatibility with a single codebase for Android and iOS.

4.2 Quality Goals

- All user data must persist correctly across app restarts without data loss.
- UI must respond correctly to all interaction states (enrolled, in-progress, completed).
- Form validation must prevent invalid profile names from being saved.
- Theme changes must apply instantly app-wide without requiring a restart.
- The app must launch within 2 seconds on a standard Android/iOS device.